

Design and comparison of the handling performance of different electric vehicle layouts

Proc IMechE Part D:
J Automobile Engineering
2014, Vol 228(2) 218–232
© IMechE 2013
Reprints and permissions:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/0954407013506569
pid.sagepub.com


Leonardo De Novellis, Aldo Sorniotti and Patrick Gruber

Abstract

In contrast with conventional vehicles driven by an internal-combustion engine, the number of motors in fully electric cars is not fixed. A variety of architectural solutions, including from one to four individually controlled electric drive units, is possible and opens up new avenues in the design of vehicle characteristics. In particular, individual control of multiple electric powertrains promises to enhance the handling performance in steady-state and dynamic conditions. For the analysis and selection of the best electric powertrain layout based on the expected vehicle characteristics and performance, new analytical tools and metrics are required. This article presents and demonstrates a novel offline procedure for the design of the feedforward control action of the vehicle dynamics controller of a fully electric vehicle and three performance indicators for the objective comparison of the handling potential of alternative electric powertrain layouts. The results demonstrate that the proposed offline routine allows the desired understeer characteristics to be achieved with any of the investigated vehicle configurations, in traction and braking conditions. With respect to linear handling characteristics, the simulations indicate that the influence of torque-vectoring is independent of the location of the controlled axles (front or rear) and is considerably affected by the number of controlled axles.

Keywords

Electric vehicles, torque vectoring, optimisation, yaw moment

Date received: 2 May 2013; accepted: 2 September 2013

Introduction

In recent years, significant improvements have been accomplished in the design of energy storage units and electric motors with high power density, high energy density and high efficiency,^{1,2} making fully electric vehicles (FEVs) an increasingly viable option for personal mobility. Current electric vehicle research is investigating different powertrain configurations constituted by one, two, three or four electric motors (Figure 1) with different performances in terms of vehicle dynamics and energy-saving targets.^{3,4}

In relation to vehicle dynamics, FEVs have the potential to achieve hitherto impossible levels of handling qualities for road vehicles, because of the very precise controllability of electric motors. In particular, advanced motor torque modulation strategies based on the combination of front-to-rear and left-to-right wheel-torque distributions (i.e. torque-vectoring (TV)) are being developed for the implementation of novel yaw rate and side-slip control algorithms and for the enhancement of brake energy recuperation, anti-lock braking system and traction control system

functions.^{5,6} The desired cornering characteristics of the vehicle can be designed primarily through a torque-vectoring control algorithm rather than through the traditional hardware-based chassis parameters such as mass distribution and suspension elastokinematics.

The increase in the vehicle configuration options presents also a challenge for electric vehicle designers to select the best architectural solution for specific vehicle design requirements. As recently pointed out by Crolla and Cao,⁷ ‘despite the significant volume of theoretical studies of torque-vectoring on vehicle-handling control, there is no widely accepted design methodology of how to exploit it to improve vehicle handling and stability significantly’. To address this issue, novel

Department of Mechanical Engineering Sciences, Faculty of Engineering and Physical Sciences, University of Surrey, Guildford, Surrey, UK

Corresponding author:

Aldo Sorniotti, Department of Mechanical Engineering Sciences, Faculty of Engineering and Physical Sciences, University of Surrey, Guildford, Surrey, UK.

Email: a.sorniotti@surrey.ac.uk

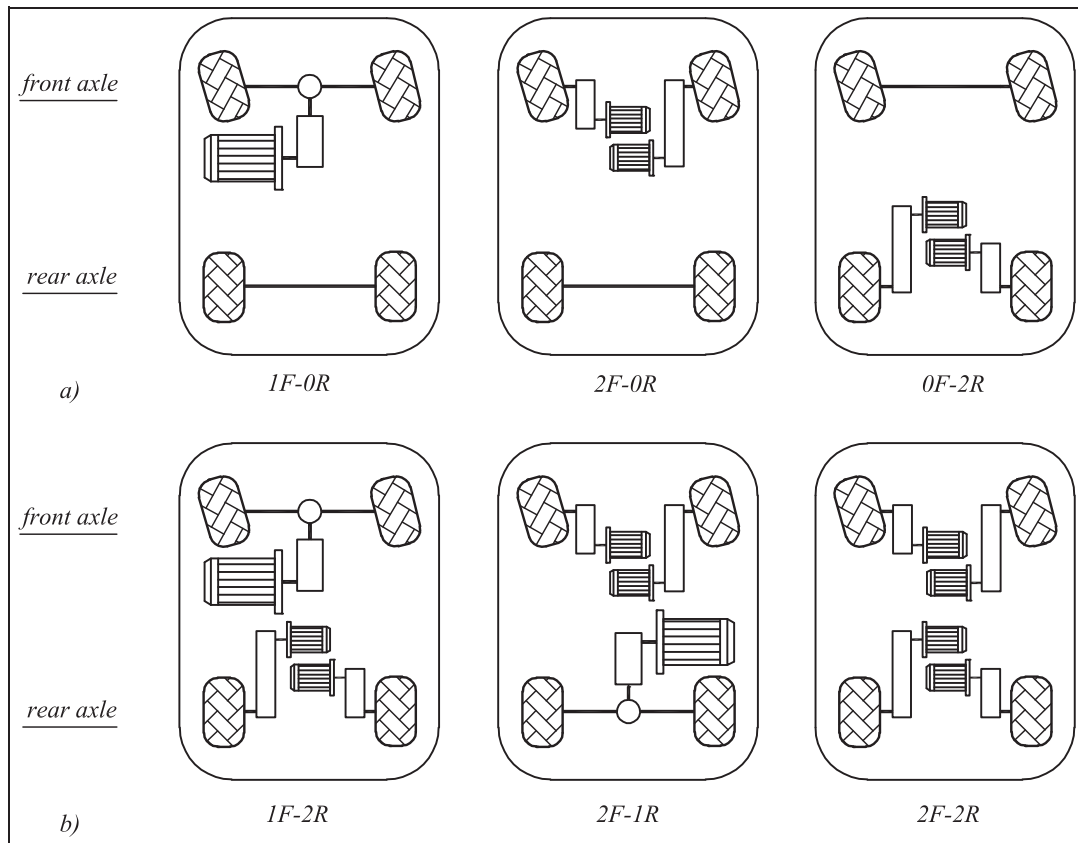


Figure 1. Examples of vehicle layouts with one to four electric powertrains. Vehicles are referred as $n_{mot}F-n_{mot}R$ where $n_{mot}F$ is the number of motors on the front axle and $n_{mot}R$ is the number of motors on the rear axle. In vehicles 1F-0R, 1F-2R and 2F-1R, the electric axle with a single motor can be equipped with a torque-vectoring differential.

analytical tools and metrics are required, which provide data for the engineer to make an informed design choice. In particular, specific torque-vectoring control methodologies for FEVs have to be developed, including the definition of the high-level targets of the vehicle's cornering response. This aspect represents one of the aims of this contribution.

In general, to avoid critical vehicle behaviour the torque-vectoring controller must be capable of continuous and smooth actuation. Current controllers adopted for conventional direct yaw moment control in production vehicles are not designed to do so as they are based on the actuation of the friction brakes when an emergency condition is detected, i.e. when the offsets between the reference and the measured (or estimated) dynamic parameters (the yaw rate r and the side-slip angle β) of the vehicle go beyond assigned thresholds.⁸ Continuous action through the integration of brake-by-wire and steer-by wire systems has been proposed to improve vehicle handling;^{9,10} in these systems, the control algorithm is based on the integrated control of active front steering and direct yaw moment control, which can be rule-based⁹ or model-based.¹⁰ However, by relying on friction brake actuation, controller interventions will reduce the vehicle speed and, thus, can be disruptive in terms of driving comfort. Also, four-wheel-steering systems^{11,12} allow improvements in yaw rate control, but they are capable of reducing the

variation in the vehicle's dynamic response induced by the longitudinal dynamics only for low values¹³ of β .

Figure 2 shows the simplified block diagram of the control structure of a torque-vectoring algorithm (which is also adopted here) suitable for FEVs, consisting of a reference yaw rate generator, a yaw moment controller, a wheel torque control allocation algorithm and a low-level controller for individual wheel slip control. In particular, the yaw moment controller is characterised by the following two parts:

- a feedforward part, which generates a reference yaw moment according to the dynamic objectives of the vehicle such as the tracking of a set of target understeer characteristics (see further discussion below);
- a feedback part, based on the difference between a reference yaw rate (output by the reference yaw rate generator in Figure 2) and the actual yaw rate, which compensates the disturbances due to system uncertainties and transient inputs but should provide limited contribution for low-steering-wheel-rate conditions.

Very different and well-known control techniques can be used for the feedback yaw moment control of vehicle dynamics. For example, regulators based on the Riccati equation,^{8,14} sliding-mode controllers¹⁵ and

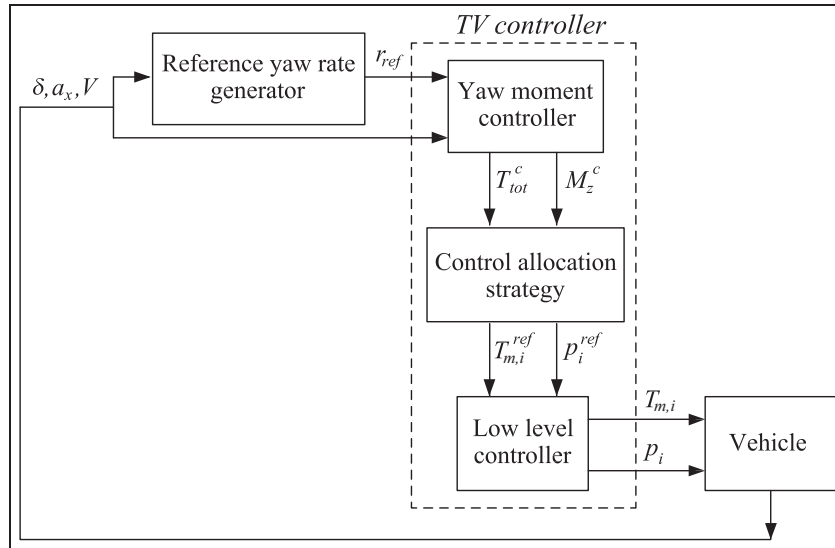


Figure 2. Control structure of the torque-vectoring algorithm for the FEVs used in the simulations.
TV: torque-vectoring.

model predictive controllers¹⁶ have been proposed in the literature and have already been applied in vehicle stability control systems based on friction brake actuation. The wheel torque distribution for the achievement of the reference yaw moment (the control allocation strategy of Figure 2) can be implemented through either rule-based algorithms¹⁷ or the application of optimal control theory.¹⁸ In order to simulate and assess the performance of the feedback part of the torque-vectoring controller, specific manoeuvres have to be simulated in the time domain.¹⁹

In contrast, the feedforward part can be tested and, hence, designed without the need for simulations in the time domain. Instead, an optimisation procedure is required to achieve the best results, which does not yet exist in the literature.

This article is an account of the development of such a novel optimisation procedure for the design of the cornering response of the FEV through the offline computation of the feedforward part of the yaw moment controller and the evaluation of its actual feasibility in terms of the vehicle's cornering response and electric drivetrain characteristics. The procedure is based on an objective function formulated according to energy efficiency criteria and constrained by a reference quasi-static vehicle-handling performance.¹¹ In addition, to complement the procedure as a powerful design tool, the paper proposes three novel performance indicators which facilitate a quick comparison of different electric vehicle architectures in terms of achievable handling characteristics.

Vehicle modelling and validation

To minimise the computational cost for running the optimisation procedure, a quasi-static vehicle model is developed and presented in the following section on vehicle models. This model is verified against results obtained

with a more advanced simulation model in the time domain. To ensure the accuracy of both vehicle models the simulation results are validated against experimental measurements, as described in the subsequent section.

Vehicle models

The optimisation procedure for the model-based derivation of the feedforward control action is applied to an eight-degree-of-freedom vehicle model. To avoid numerical forward time integration of the equations of motion, the vehicle model is based on a quasi-static approximation which assumes the yaw acceleration $\dot{r}$ and the side-slip rate $\dot{\beta}$ to be zero. The degrees of freedom of the model are the longitudinal, lateral, roll and yaw motions of the vehicle body and the rotations of the four wheels. Assuming a flat road surface and a small vehicle side-slip angle, the vehicle body dynamics are described by the longitudinal force balance equation (neglecting the equivalent mass of the wheel's rotational inertia, included in the equations describing the drivetrain dynamics)

$$\sum_{i=1}^4 F_{x_i} \cos \delta_{w_i} + \sum_{i=1}^4 F_{y_i} \sin \delta_{w_i} - F_{drag} = m(\dot{u} - rV\beta) \quad (1)$$

the lateral force balance equation

$$\sum_{i=1}^4 F_{x_i} \sin \delta_{w_i} + \sum_{i=1}^4 F_{y_i} \cos \delta_{w_i} = m(\dot{V}\beta + ur) \quad (2)$$

the yaw moment balance equation

$$M_z = \sum_{i=1}^4 F_{x_i} \cos \delta_{w_i} y_i + \sum_{i=1}^4 F_{x_i} \sin \delta_{w_i} x_i + \sum_{i=1}^4 F_{y_i} \cos \delta_{w_i} x_i + \sum_{i=1}^4 F_{y_i} \sin \delta_{w_i} y_i + \sum_{i=1}^4 M_{z_i} \quad (3)$$

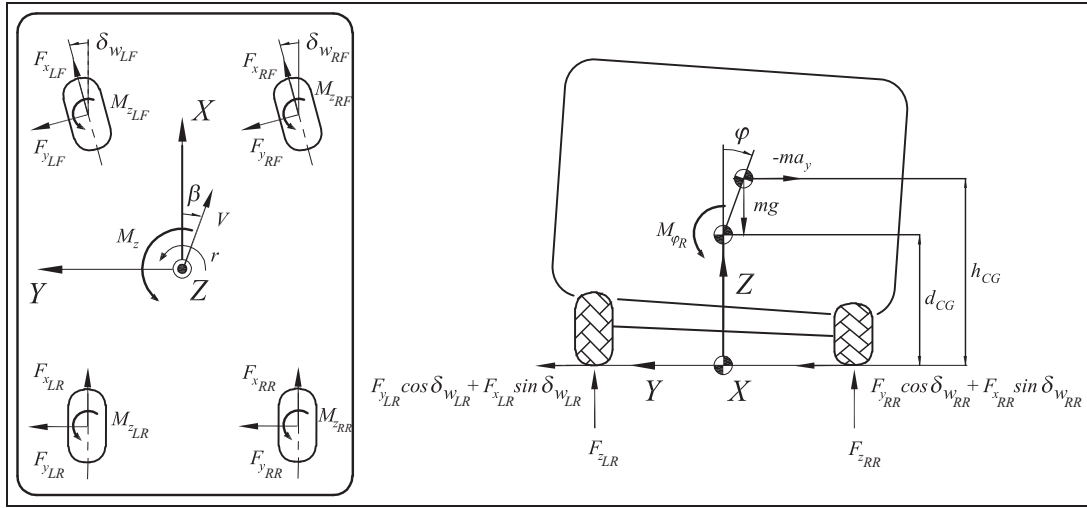


Figure 3. Top and rear views of the free-body diagram of the vehicle based on the ISO vehicle reference system.²⁰ In the rear view (Y-Z plane), only quantities related to the rear axle are represented, together with the vehicle's roll centre at the centre of mass.

and the roll moment balance equation

$$\begin{aligned}
 & m(\dot{V}\beta + ur)(h_{CG} - d_{CG}) \cos \varphi + mg(h_{CG} - d_{CG}) \sin \varphi \\
 & - \left(\sum_{i=1}^2 F_{x_{iF}} \sin \delta_{w_{iF}} + \sum_{i=1}^2 F_{y_{iF}} \cos \delta_{w_{iF}} \right) (d_F - d_{CG}) \\
 & - \left(\sum_{i=1}^2 F_{x_{iR}} \sin \delta_{w_{iR}} + \sum_{i=1}^2 F_{y_{iR}} \cos \delta_{w_{iR}} \right) \\
 & (d_R - d_{CG}) = M_{\varphi_F} + M_{\varphi_R}
 \end{aligned} \quad (4)$$

where the subscripts F and R refer to the front axle and the rear axle respectively, u is the component of the velocity V of the centre of gravity of the vehicle along the longitudinal axis of the vehicle reference system, F_{x_i} , F_{y_i} and M_{z_i} are the longitudinal force, the lateral force and the self-aligning moment respectively for the i th tyre, evaluated in the tyre reference system, and M_z is the yaw moment required to maintain the vehicle in trimmed conditions according to the quasi-static approach. The steering angle δ_{w_i} of each wheel takes into account the kinematic contributions due to the suspension and steering-system design and the compliance effect induced by the load applied to the wheel. x_i and y_i are the longitudinal distance and lateral distance respectively between each tyre and the vehicle's centre of gravity; also these parameters are subject to variations depending on the elastokinematics of the suspension. h_{CG} is the height of the vehicle's centre of gravity, and d_{CG} , d_F and d_R are the heights of the roll axis measured at the vehicle's centre of gravity, the front suspension and the rear suspension respectively. The front anti-roll moment M_{φ_F} and rear anti-roll moment M_{φ_R} are expressed in the form of non-linear look-up tables taking into account only the roll stiffness contribution as the roll rate is considered to be zero ($\dot{\varphi} = 0$) in the employed quasi-static approach. F_{drag} is the

aerodynamic drag force. Figure 3 shows the corresponding free-body diagrams of the vehicle.

The Pacejka '96 tyre model was employed to evaluate F_{x_i} , F_{y_i} and M_{z_i} as functions of the longitudinal slip σ_i , the slip angle α_i , the camber angle γ_i , the tyre-road friction coefficient and the vertical load F_{z_i} . F_{z_i} on the i th front (F) wheel or rear (R) wheel is given by

$$\begin{aligned}
 F_{z_i} = F_{z0i} \pm [F_{drag} + m(\dot{u} - rV\beta)] \frac{h_{CG}}{2L} \\
 \pm \frac{\sum_{j=1}^2 F_{x_{jF/R}} \sin \delta_{w_{jF/R}} + \sum_{j=1}^2 F_{y_{jF/R}} \cos \delta_{w_{jF/R}}}{\sum_{j=1}^2 |y_{jF/R}|} d_{F/R} \\
 \pm \frac{M_{\varphi_{F/R}}}{\sum_{j=1}^2 |y_{jF/R}|}
 \end{aligned} \quad (5)$$

where the summations $\sum_{F/R}$ are applied to the two wheels of the same axle, F_{z0i} is the static vertical load on the tyre and L is the vehicle wheelbase.

The wheel moment balance equations have the structure

$$\begin{aligned}
 T_i - T_{FB_i} - F_{x_i} R_i - F_{z_i} R_i (f_0 + f_1 \omega_i R_{roll_i} + f_2 \omega_i^2 R_{roll_i}^2) \\
 - J_{W_i} \dot{\omega}_i = 0
 \end{aligned} \quad (6)$$

where T_i and T_{FB_i} are the electric drivetrain torque at the wheel and the friction brake torque respectively, R_i and R_{roll_i} are the laden radius and the rolling radius respectively of the tyre and J_{W_i} is the moment of inertia of the wheel. The factors f_0 , f_1 and f_2 represent the components of the rolling resistance coefficient and $\dot{\omega}_i$ is the angular acceleration of the wheel, which is expressed as a function of the slip ratio and the velocity component V_{x_i} of the i th wheel hub along the x axis of the tyre, according to

$$\begin{aligned}\dot{\omega}_i &= \frac{d}{dt} \left[\frac{V_{x_i}}{R_{roll_i}} (\sigma_i + 1) \right] \\ &= \frac{a_{x_i}}{R_{roll_i}} (\sigma_i + 1) + \frac{V_{x_i}}{R_{roll_i}} \dot{\sigma}_i, \quad i = 1, \dots, 4\end{aligned}\quad (7)$$

where the time derivative $\dot{\sigma}_i$ of the slip ratio of the tyre can be neglected owing to the quasi-static approach. The set of algebraic equations (1) to (7) is completed with additional equations related to the kinematic relationships for the evaluation of the slip angles $\alpha_i = f_\alpha(\beta, \delta_{w_i}, r, F_{x_i}, F_{y_i})$ and longitudinal slip ratios $\sigma_i = f_\sigma(V, \omega_i, \delta_{w_i}, r, F_{x_i}, F_{y_i})$, where the explicit definitions of f_α and f_σ , which take into account the suspension compliances, are omitted for brevity. The wheel torque distribution can be expressed as $T_i = f_{T_i}(T_{TOT})$, where $T_{TOT} = \sum_{k=1}^n T_k$ and f_{T_i} is the torque distribution criterion.

The difference between the alternative electric drivetrain layouts in Figure 1 is included in the dynamic equation linking the wheel torque T_i to the electric motor torque T_{m_i} . For example, with individual wheel drivetrains consisting of an on-board electric motor drive, an on-board two-stage single-speed transmission and a half-shaft with constant-velocity joints, the relevant equation is

$$T_{m_i} - \frac{T_i}{\tau_{1_i} \tau_{2_i} \eta_{1_i} \eta_{2_i} \eta_{CV1_i} \eta_{CV2_i}} - J_{eq_i} \dot{\omega}_{m_i} = 0 \quad (8)$$

where τ_{1_i} and τ_{2_i} are the reduction ratios of each of the two stages of the single-speed transmission, η_{1_i} and η_{2_i} are the equivalent efficiencies of the transmission stages, and η_{CV1_i} and η_{CV2_i} are the efficiencies of the constant-velocity joints located at the two ends of the half-shaft. J_{eq_i} includes the inertial contributions due to the individual components of the drivetrain, namely

$$\begin{aligned}J_{eq_i} &= J_{mot_i} + J_{1_i} + \frac{J_{2_i}}{\tau_{1_i}^2 \eta_{1_i}} + \frac{J_{3_i}}{\tau_{1_i}^2 \tau_{2_i}^2 \eta_{1_i} \eta_{2_i}} \\ &\quad + \frac{J_{HS_i}}{\tau_{1_i}^2 \tau_{2_i}^2 \eta_{1_i} \eta_{2_i} \eta_{CV1_i} \eta_{CV2_i}}\end{aligned}\quad (9)$$

where J_{1_i} , J_{2_i} and J_{3_i} are the moments of inertia of the primary shaft, the secondary shaft and the output shaft respectively of the single-speed transmission and J_{HS_i} is the moment of inertia of the half-shaft.

For the case of a single electric motor on a driven axle (vehicles 1F-0R, 1F-2R and 2F-1R in Figure 1), the model of a torque-vectoring differential was included, which uses multi-plate clutch packs to distribute the torque between the left and right wheels. The particular model is adopted from the work by Sawase et al.,²¹ which simulates an overdriven torque-vectoring differential allowing the possibility of a torque bias also towards the faster wheel of the axle. The associated power losses are estimated from the product of the differential torque output and the slip velocity of the differential clutch pack.

To calculate the input power to the electric powertrain, the electric motor drives are modelled with

efficiency maps that are functions of the primary operating variables, i.e. the torque, the speed, the input voltage and the operating temperature of the motor. Also, a realistic representation of the vehicle battery is provided by a dynamic battery model which is based on the approach outlined by Gao.²²

For verification, the results obtained with the quasi-static vehicle model are compared with those computed with a more detailed vehicle model in the time domain that was implemented in the vehicle dynamics simulation software IPG CarMaker²³ and validated (see the section on the experimental validation of the models). To include the six different electric powertrain layouts shown in Figure 1, a MATLAB/Simulink dynamic model was integrated in the IPG CarMaker model. With this modelling approach, the first-order dynamics of the drivetrain were taken into account, thus considering the torsion dynamics of the half-shafts, the plays within the drivetrain and the relaxation length of the tyre.²⁴

Experimental validation of the models

Experiments were carried out with a vehicle demonstrator (a front-wheel-drive sport utility vehicle) at the Lommel proving ground (Belgium).

In accordance with the standards ISO 4138:2012²⁵ and ISO 7401:2011,²⁶ skid-pad and step-steer manoeuvres were performed under a wide variety of operating conditions, i.e. selected gear, trajectory radius and vehicle velocity.

For model validation, the experimentally measured time histories of the steering-wheel angle and the vehicle speed were provided as inputs to the IPG CarMaker simulator. As indicated by Figures 4 and 5, during the ramp-steer and step-steer manoeuvres the yaw rate response predicted by the IPG CarMaker model matches the experimental measurements well. The understeer characteristics and the side-slip angle characteristics, as functions of the vehicle's lateral acceleration for the test vehicle, the IPG CarMaker simulator and the quasi-static model, are compared in Figure 6

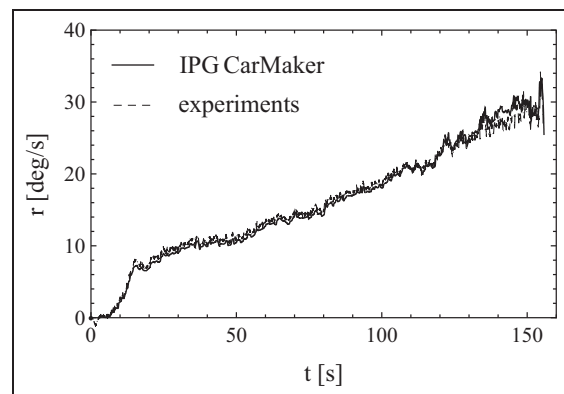


Figure 4. Yaw rate response in slowly varying conditions (skid-pad test): comparison between the results obtained from the experiments and the IPG CarMaker simulation model.

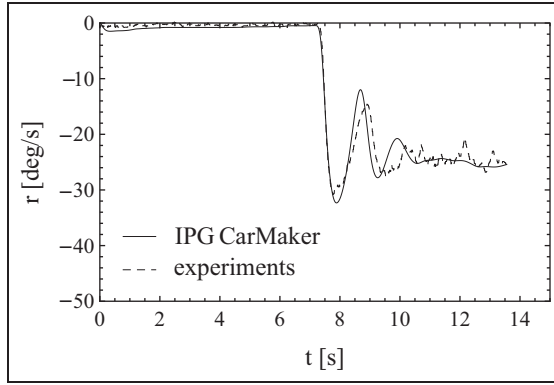


Figure 5. Yaw rate response in transient conditions (step-steer test): comparison between the results obtained from the experiments and the IPG CarMaker simulation model.

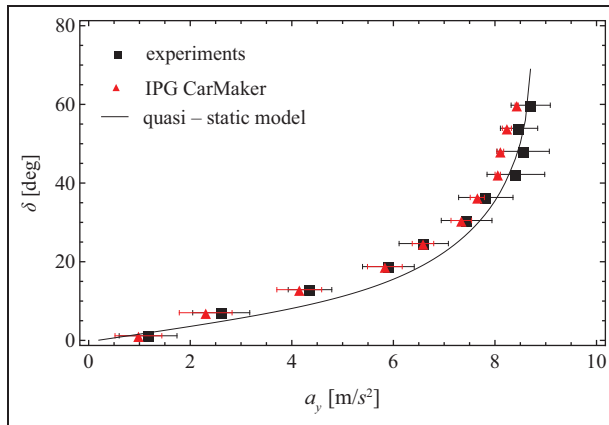


Figure 6. Understeer characteristics, showing the steering-wheel angle δ as a function of the lateral acceleration a_y for a trajectory radius of 60 m: comparison between the experimental results, the quasi-static model and the IPG CarMaker simulation model predictions.

and Figure 7 respectively. The offsets related to the kinematic values of δ and β were subtracted from the actual values of the parameters.²⁷ The horizontal bars, shown for the experimental and the IPG CarMaker model results, indicate the range of variations (in terms of the standard deviation with respect to the mean value) of the lateral acceleration in the time domain due to steering-wheel angle oscillations that were measured during the particular manoeuvres. Owing to the good match with the experimental results, the dynamic and quasi-static models can be assumed to simulate accurately and reliably the linear and non-linear vehicle responses and can be adopted as predictive tools for evaluation of the handling responses of different FEV layouts.

The offline optimisation procedure for the design of the torque-vectoring controller

Torque vectoring and vehicle understeer

By varying the distribution of the traction or braking torques (requested by the driver) between the driven

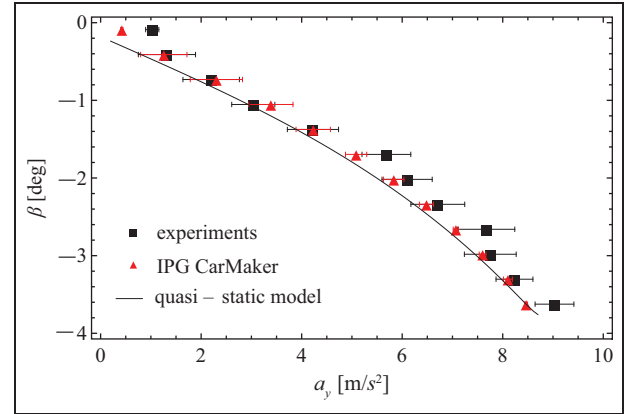


Figure 7. Side-slip angle characteristics, showing the side-slip angle β as a function of the lateral acceleration a_y for a trajectory radius of 60 m: comparison between the experimental results, the quasi-static model and the IPG CarMaker simulation model predictions.

wheels and, thus, by influencing the vehicle yaw moment M_z , the understeer behaviour of a vehicle can be significantly modified.

For instance, in trimmed conditions and disregarding the self-aligning moments of the tyre, equation (3) becomes $M_z \approx M_{z,F_x} + M_{z,F_y} = 0$, where M_{z,F_x} is the yaw moment contribution due to the longitudinal tyre forces and M_{z,F_y} is the yaw moment contribution due to the lateral tyre forces. As the intervention of the torque-vectoring controller generates a difference between the longitudinal forces of the left and the right wheels, $M_{z,F_x} \neq 0$ and, hence, $M_{z,F_y} = -M_{z,F_x}$. The condition $M_{z,F_y} \neq 0$ implies a variation in the lateral forces on the front axle and the rear axle in comparison with the vehicle without torque vectoring. In trimmed conditions, without torque vectoring, $M_{z,F_x} \approx 0$ (M_{z,F_x} is not exactly zero because of the marginal difference between the left front and right front steering angles), from which it follows that $M_{z,F_y} \approx 0$. As a consequence, if during traction the longitudinal tyre forces are larger on the outer side of the corner, the lateral force on the front axle in trimmed conditions will be lower than for the vehicle without torque-vectoring at the same lateral acceleration, and the lateral force on the rear axle will be larger. As the lateral tyre forces relate to the tyre slip angles, the front and rear slip angles will change with respect to the vehicle without torque vectoring. The level of vehicle understeer depends on the difference between the average front and rear slip angles of each axle; therefore the employment of torque-vectoring control allows the understeer characteristic of the vehicle to change in trimmed conditions.

The complexity of this relationship is further increased by the interaction between the longitudinal tyre forces and the lateral tyre forces, according to the friction ellipse. This interaction makes the vehicle response sensitive to the front-to-rear torque-vectoring distribution, i.e. the front and rear axles can contribute differently to the generation of M_{z,F_x} . Therefore, the

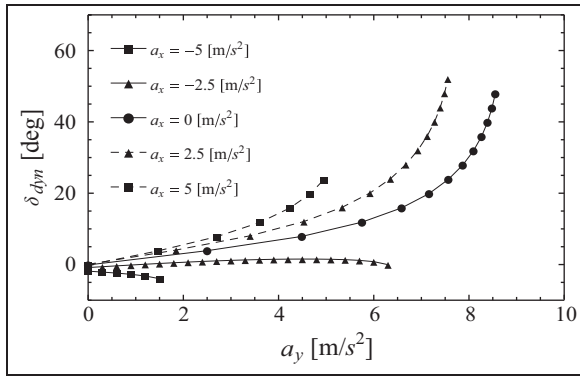


Figure 8. Understeer characteristics at a longitudinal velocity of 90 km/h for different values of the longitudinal acceleration a_x , ranging from -5 m/s^2 to 5 m/s^2 in steps of 2.5 m/s^2 , for a vehicle with a constant wheel torque distribution in traction and braking (vehicle layout 2F–2R of Figure 1). The simulation results were obtained with the quasi-static model of the second section.

same value of M_{z,F_x} can produce different understeer characteristics in trimmed conditions, especially when the lateral acceleration approaches its limit.

The design specifications of the torque-vectoring controller

As indicated by Figure 8, the understeer characteristics in traction and braking conditions can vary significantly owing to the effect of the longitudinal load transfer. This variation leads to rather different vehicle turn-in behaviours which may not be predicted by the normal driver and could lead to critical driving manoeuvres. To achieve a steering behaviour that is less influenced by a_x , the feedforward part of the torque-vectoring controller can be designed for specific handling targets.

For this study, four realistic vehicle-handling targets in comparison with the same vehicle with a constant wheel torque distribution were set for trimmed conditions (i.e. $M_z = 0$). The objectives were chosen to achieve a vehicle that is predictable and easy to control to enhance vehicle safety and yet can be set up to improve agility to make the car feel sporty and direct. The handling targets are as follows:

- a reduction in the understeer gradient $K_U = \partial \delta_{dyn} / \partial a_y$ (where δ_{dyn} is the dynamic steering-wheel angle and a_y is the lateral acceleration) in the linear part of the understeer characteristic (i.e. the part of the understeer characteristic for which the variation in K_U is within an assigned limited percentage threshold) for $a_x = 0$;
- an extension of the area of linearity of the understeer characteristic at $a_x = 0$;
- an increase in the maximum value $a_{y,MAX}$ of the vehicle's lateral acceleration;
- a reduction in the variation in the understeer characteristic as a function of a_x (induced by traction and braking).

Targets (a) to (c) are realised through an increase in the torque on the wheels on the outer side of the corner and a decrease in the wheel torque on the inner side. Target (d) can be achieved, for example, with torque-vectoring strategies such as those proposed by Shibata and Shibahata²⁸ and Tanaka et al.,²⁹ which are based on the traction forces F_{x_i} distributed proportionally to the vertical tyre load F_{z_i} . However, the benefit of these strategies is limited as the achievable extent of the reduction in the spread of the understeer characteristics cannot be predicted a priori. Moreover, the wheel load dependence strategy is ineffective in achieving targets (a) to (c) as it is not explicitly based on a reference understeer characteristic.

Hence, in order to achieve objectives (a) to (d) simultaneously, a novel procedure for the offline design of the feedforward torque-vectoring control action was developed.

The optimisation-based design of the feedforward part of the torque-vectoring controller

The problem of designing the feedforward part of the torque-vectoring controller is addressed as an optimisation problem where a suitable objective function has to be minimised, taking into account physical constraints. The developed algorithm consists of three steps as discussed in the following three sections.

Definition of the set of reference understeer characteristics. In order to quantify the handling targets defined in the section on the design specifications of the torque-vectoring controller, a formulation for the reference understeer characteristic is required. Therefore, the target value of the dynamic steering-wheel angle δ_{dyn} (i.e. the difference between the actual steering-wheel angle and its kinematic value) as a function of the lateral acceleration can be defined for the relevant range of a_x . Based on the correlation with experimental data from different vehicles, a suitable analytical formulation was found and is given by

$$a_y = \frac{1}{K_U} \delta_{dyn} \text{ if } \delta_{dyn} < a_y^* K_U \quad (10)$$

$$a_y = a_{y,MAX} + \left(a_y^* - a_{y,MAX} \right) e^{\frac{(a_y^* K_U - \delta_{dyn})}{(a_{y,MAX} - a_y^*) K_U}} \text{ if } \delta_{dyn} \geq a_y^* K_U \quad (11)$$

Equation (10) defines the linear part of the reference understeer characteristic³⁰ and equation (11) describes the non-linear part of the understeer characteristic, which arises from tyre saturation. The resulting function makes use of three variables that correspond to the previously defined handling targets for the cornering behaviour of the vehicle: the understeer gradient $K_U(a_x)$, the linear limit acceleration threshold $a_y^*(a_x)$ and $a_{y,MAX}(a_x)$, which is the maximum lateral acceleration achievable by the vehicle.

Definition of the problem constraints. The physical limits of the fully electric vehicle are taken into account by setting the system constraints in terms of the maximum electric motor torque and power characteristics, and the peak power of the battery pack. The combination of the physical constraints, the equations of the quasi-static model and equations (10) and (11) represent a set of equality constraints that do not fully constrain a system with two driven axles and left-to-right torque-vectoring within at least one of them. As a consequence, the minimisation of a secondary cost function is required as discussed in the next section on the definition of the objective function.

Definition of the objective function. For this study, an objective function J_P related to the energy efficiency of the vehicle is used. It is based on the overall input power P_{TOT} of the electric powertrain depending on the contributions P_{input_i} of each motor drive and is given by

$$J_P = P_{TOT} = \sum_{i=1}^{n_{mot}} P_{input_i} \quad (12)$$

Hence, M_{z,F_x} and the optimal electric motor torque distribution will be evaluated through minimisation of J_P . The algorithm which has given robust solutions of the optimisation problem for a variety of vehicle layouts and electric motor efficiency maps (usually the most critical element in the procedure³¹) is the interior-reflective Newton method.³²

Results obtained with the optimisation procedure

The output of the optimisation procedure is a look-up table for the wheel torque distribution as a function of the steering-wheel angle, the vehicle speed and the accelerator and brake pedal demands. The look-up table forms the feedforward part of the reference yaw moment M_{z,F_x} which is added to the feedback part (zero in quasi-static conditions), as shown in the control structure in Figure 2. Within an online implementation, M_{z,F_x} is sent to a control allocation algorithm based on the same objective function J_P adopted within the offline optimisation method. This scheme has the following benefits:

- design of the feedforward control action based on consistent control targets for any operating condition (this is not possible with simulations or experimental tests in the time domain); the set of reference understeer characteristics is also converted into the corresponding reference yaw rate look-up table for the feedback part of the controller;
- very quick design and critical comparison of alternative sets of achievable feedforward control actions, as the tool is computationally efficient owing to the quasi-static modelling approach;
- the a-priori comparison of different wheel torque control allocation techniques (by changing the

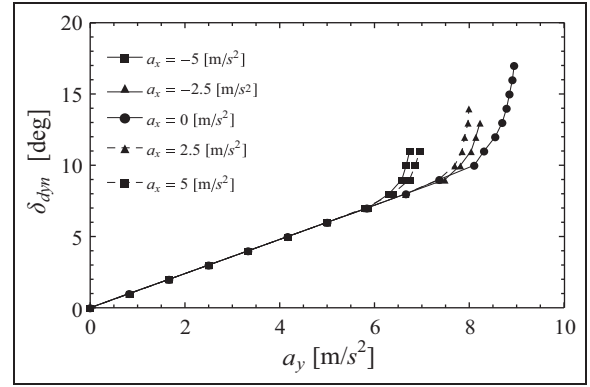


Figure 9. Understeer characteristics at a longitudinal velocity of 90 km/h for different values of the longitudinal acceleration a_x , ranging from -5 m/s^2 to 5 m/s^2 in steps of 2.5 m/s^2 , for a vehicle with the torque-vectoring distribution designed through the optimisation tool (vehicle layout 2F–2R of Figure 1).

objective function of the offline optimisation), without the limitations and simplifications deriving from the actual online numerical implementation of the algorithms;

- the a-posteriori verification of the performance of the online control allocation algorithm through its comparison with the ideal output of the offline optimisation procedure.

As an example of the obtained results, Figure 9 plots the understeer characteristics for the same vehicle parameter set used for Figure 8 but adopts a feedforward controller designed with the optimisation procedure described above. The vehicle response is fundamentally transformed from Figure 8; it is consistent with the outlined high-level objectives of the torque-vectoring controller (see the section on the design specifications of the torque-vectoring controller) and independent of the accelerator and the brake pedal inputs, apart from the limited non-linear region.

The practical impact of defining a reference set of understeer characteristics is shown in Figure 10. The graph compares the simulation results obtained with the IPG CarMaker/Simulink model in terms of the response in the time domain of the vehicles of Figures 8 and 9 during a tip-in manoeuvre (i.e. fast application of a significant accelerator pedal input) carried out in cornering conditions (from the same initial a_y). Compared with the vehicle with a fixed wheel torque distribution, the vehicle with torque-vectoring control does not show substantial variations in its yaw rate response, yielding a major benefit in terms of the vehicle safety and the driver's effort reduction.

Evaluation metrics for different vehicle dynamic performances

The two key challenges in the design of an electric vehicle layout with torque-vectoring capabilities are to

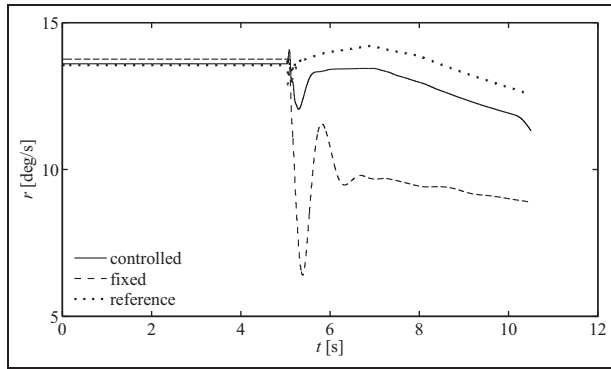


Figure 10. Tip-in manoeuvre in cornering conditions (vehicle layout 2F-2R of Figure 1): yaw rate response of the torque-vectoring-controlled vehicle (solid curve labelled controlled), reference yaw rate response according to the set of reference understeer characteristics of Figure 9 (dotted curve labelled reference) and yaw rate response of the vehicle with fixed wheel torque distribution (dashed curve labelled fixed).

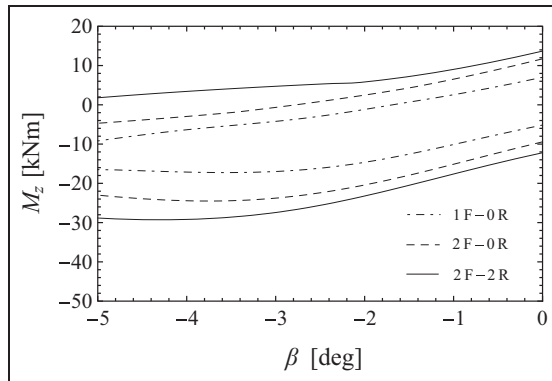


Figure 11. Comparison of the boundaries of $M_{z,MAX}$ and $M_{z,MIN}$ for the vehicle layouts 1F-0R, 2F-0R and 2F-2R of Figure 1 (between $\beta_{MIN} = -5^\circ$ and $\beta_{MAX} = 0^\circ$), for a range of steering wheel angles δ between -100° and 100° .

understand, first, whether to adopt a two-wheel-drive (2WD) layout or a four-wheel-drive (4WD) layout and, second, whether to adopt a torque-vectoring differential or two individually controlled motor drives for each driven axle. These choices are functions of the expected target understeer characteristics for the specific application. This section outlines how the optimisation algorithm previously described (in the third section) can provide necessary information to address these two challenges. In particular, to facilitate an objective evaluation and comparison of the simulation results, three handling performance indicators I_1 , I_2 and I_3 are introduced.

Limit of linear vehicle behaviour (indicator I_1)

In order to evaluate the cornering capability of the vehicle within the linear response region, indicator I_1 is proposed. It is based on the maximum value of a_y , here

called $a_{y,MAX}^*$, for which the vehicle can maintain a target constant K_U at the considered value of a_x . $a_{y,MAX}^*$ is derived by employing the algorithm described in the section on the optimisation-based design of the feedforward part of the torque-vectoring controller through maximisation of the objective function (which replaces that of equation (12))

$$J_A = \sum F_{x_i} \sin \delta_{w_i} + \sum F_{y_i} \cos \delta_{w_i} \quad (13)$$

with the main constraints being the reference linear understeer characteristics and $M_z = 0$. Since $a_{y,MAX}^*$ is a function of a_x , the area of the region of the a_x - a_y plan (or g - g diagram²⁷) within the boundaries of the linear operating response of the controlled vehicle is proposed as the performance indicator I_1 which is given by

$$I_1 = \int_{a_{x1}}^{a_{x2}} a_{y,MAX}^*(a_x) da_x \quad (14)$$

From the viewpoint of driving experience, a high I_1 value is desirable as it provides a feeling of consistency to the driver and enhances the drivers' perceptions in terms of the vehicle agility and the 'fun-to-drive' aspect. In the area covered by $a_{y,MAX}^*(a_x)$ and quantified by I_1 , the user will experience an 'easy-to-drive' vehicle.

Maximum lateral acceleration of the vehicle (indicator I_2)

To examine the ultimate cornering capability of the vehicle in traction and braking conditions, the performance indicator I_2 is defined on the basis of the area covered by the graph of the achievable maximum lateral acceleration³³ $a_{y,MAX}(a_x)$. In order to compute the $a_{y,MAX}(a_x)$ characteristics, the optimisation algorithm (in the third section) is used to maximise the cost function J_A (equation (13)), while considering the main constraint $M_z = 0$ and without any condition on the steering-wheel angle. Over the range of the considered longitudinal accelerations, I_2 is evaluated as

$$I_2 = \int_{a_{x1}}^{a_{x2}} a_{y,MAX}(a_x) da_x \quad (15)$$

In terms of the driving experience, I_2 will correlate with the test drivers' perception of the vehicle on a test track and will be mainly useful for high-performance vehicles.

Vehicle controllability (indicator I_3)

The constraint to operate on an assigned set of reference understeer characteristics in quasi-static conditions is not sufficient to provide a consistent enhancement of the vehicle-handling performance. In fact, the reduction in the vehicle understeer in traction (in order to improve the 'fun-to-drive' aspect) usually implies a reduction in the stability in a significant portion of the

vehicle's operating conditions. This interrelationship is due to the increase in the lateral acceleration for the same value of the steering-wheel angle, which causes a larger side-slip angle and yaw rate oscillations in transient conditions. As a consequence, the controller (especially in its feedback contribution) must be capable of a significant dynamic correction of the vehicle response in order to provide the expected dynamic qualities under all possible driving conditions, including transients. Vehicle controllability during steering inputs can be estimated through evaluation of the top boundary $M_{z,MAX}$ and the bottom boundary $M_{z,MIN}$ of the achievable yaw moments as functions of the vehicle side-slip angle.¹⁵ By applying the optimisation algorithm to the quasi-static vehicle model with the cost function $J_M = M_z$ (defined by equation (3)) to be maximised and minimised $\forall \delta \in [\delta_{MIN}, \delta_{MAX}]$, the trend of $M_{z,MAX}(\beta)$ and $M_{z,MIN}(\beta)$ defining the controllability area of M_z can be derived.

Figure 11 compares the boundaries of the yaw moment plots for the vehicle layouts 1F-0R, 2F-0R and 2F-2R in conditions of constant velocity ($V = 90$ km/h). M_z is positive when it is a destabilising moment. For the case-study vehicles with two motor drives per axle, the difference in the yaw moment controllability range $M_{z,MAX} - M_{z,MIN}$ is an increasing function of $|\beta|$. For the vehicle 1F-0R, starting from $|\beta| > 3^\circ$ the yaw moment controllability range decreases. Moreover, at $|\beta| \approx 2^\circ$, $M_{z,MAX}$ becomes negative, indicating that the vehicle cannot be corrected for understeering behaviour with increasing $|\beta|$. This limitation is overcome with the 2F-2R configuration because of its positive maximum yaw moment for all tested $|\beta|$. For larger values of $|\beta|$ than those included in Figure 11, the yaw moment controllability range would decrease.

Based on the area between the top boundary and the bottom boundary (Figure 11), the performance indicator I_3 is defined. To account for the stabilising and destabilising yaw moments, I_3 is composed of two parameters $I_{3,Stab}$ and $I_{3,Dest}$ relating to positive M_z and negative M_z respectively along a range of side-slip angles $\{\beta_{MIN}, \dots, \beta_{MAX}\}$, which are given by

$$I_{3,Stab} = \int_{M_{z,MIN}}^{M_{z,MAX}} \int_{\beta_{MIN}}^{\beta_{MAX}} dM_z d\beta \quad \left| \begin{array}{l} M_z < 0 \end{array} \right. \quad (16)$$

$$I_{3,Dest} = \int_{M_{z,MIN}}^{M_{z,MAX}} \int_{\beta_{MIN}}^{\beta_{MAX}} dM_z d\beta \quad \left| \begin{array}{l} M_z > 0 \end{array} \right. \quad (17)$$

In particular, $I_{3,Stab}$ is an essential indicator of the vehicle safety margin in transient conditions and can be adopted for predicting the potential effectiveness of the stability control system: the larger $I_{3,Stab}$ becomes, the larger is the stabilisation moment that can be applied in an emergency manoeuvre in order to restore safer driving conditions.

Results

This section studies the handling performance potential of the vehicle layouts in Figure 1 on the basis of the three performance indicators defined in the previous sections. For vehicles 1F-2R and 2F-1R, the options of a torque-vectoring differential or an open differential are included for the axle driven by a single electric motor. The torque-vectoring differential, where applicable, is modelled to provide a maximum torque bias $|T_{dLeft} - T_{dRight}|$ of 800 Nm. The main vehicle parameters are listed in Table 1 in Appendix 2.

To ensure comparable results, the overall wheel drive torque characteristics of each tested powertrain configuration are kept the same as those of a reference vehicle, which is currently developed in reality. The reference vehicle has a 2F-2R architecture with four on-board switched reluctance electric motor drives that were experimentally tested to derive the relevant simulation parameters. The same characteristics with one, two or three electric drivetrains were obtained by scaling the known motor drive torque characteristics while keeping the same base speed (i.e. the speed at which the transition between the constant-torque region and the constant-power region of the electric motor occurs). As a result, all tested vehicle layouts have approximately the same longitudinal acceleration performances. The data for the scaled units in terms of the masses and the moments of inertia were estimated in collaboration with the motor manufacturer.

The results for the performance indicators I_1 , I_2 and I_3 were obtained under the hypothesis that, for deceleration conditions, the brake pressure for the friction brake system can be freely modulated between the two axles (as, for example, with an electrohydraulic brake system), but not within each axle, i.e. between the left and the right callipers (as, for instance, with a vehicle dynamics control system). For traction conditions, it is stipulated that actuation of the friction brakes is not allowed.

Figure 12 plots $a_{y,MAX}^*$ as a function of a_x for four characteristic electric vehicle layouts. The

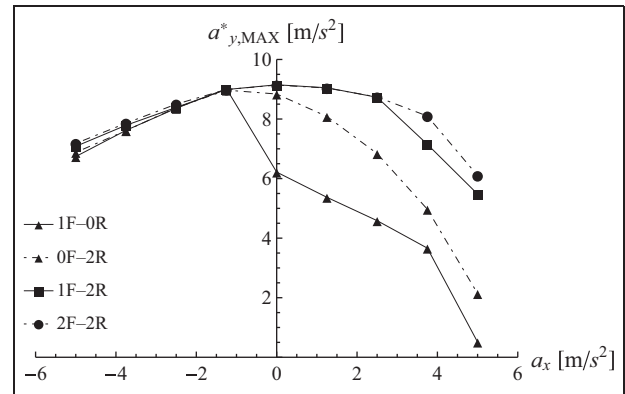


Figure 12. $a_{y,MAX}^*$ as a function of a_x , for the vehicle layouts with torque-vectoring control 1F-0R, 0F-2R, 1F-2R and 2F-2R at 90 km/h and with $K_U = 10$ deg/g. For clarity, a reduced number of data points are shown.

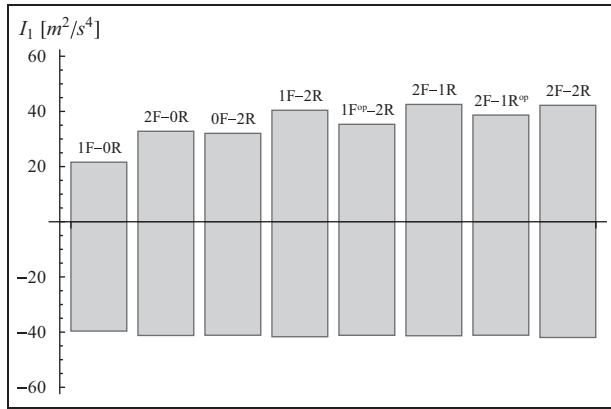


Figure 13. I_1 for the eight alternative electric vehicle layouts, at 90 km/h and with $K_U = 10$ deg/g.

corresponding performance indicators I_1 are shown in Figure 13. For the simulation, the reference value of K_U for the controlled vehicles is fixed to 10 deg/g, which is approximately two thirds of the K_U value experimentally measured for the front-wheel-drive vehicle at zero longitudinal and lateral accelerations (see the section on experimental validation of the models). In Figure 13, the results for I_1 are shown for traction conditions (top half of the diagram) and for braking conditions (bottom half of the diagram). As expected, the different electric vehicle layouts show almost identical results in braking because of their similar wheel brake torque capabilities achieved by the blending of friction braking and regenerative braking. In traction, the studied powertrain configurations exhibit different characteristics. Owing to the torque bias limitation, the vehicle with a torque-vectoring differential (1F-0R) has a significantly reduced area of possible linear vehicle responses compared with the other examined 2WD layouts. In traction, I_1 for vehicles 2F-0R and 0F-2R is approximately 30% higher than for vehicle 1F-0R. The very similar I_1 results for the rear-wheel-drive and front-wheel-drive layouts imply that the choice of drive axle does not yield advantages in terms of extension of the area of linear response for the specific case-study vehicle.

The largest I_1 values and, thus, the widest linear handling region can be achieved with the 4WD layouts. For example, compared with the rear-wheel-drive vehicle 0F-2R, the performance indicator increases by 22% in traction conditions when the 1F-2R configuration is selected. This behaviour can be expected, considering the well-known safety benefits that are already achieved with the modern 4WD systems of conventional cars. However, the proposed indicator I_1 allows the relative potential performance gain between the different layouts to be quantified.

Figure 14 and Figure 15 plot $a_{y,MAX}$ against a_x for the same four characteristic electric vehicle layouts, with and without torque-vectoring control respectively. For the case of 4WD vehicles without torque-vectoring

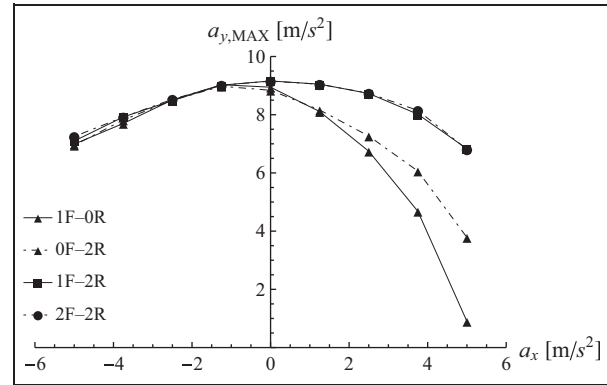


Figure 14. $a_{y,MAX}$ as a function of a_x at 90 km/h for the vehicle layouts 1F-0R, 0F-2R, 1F-2R and 2F-2R with torque-vectoring control.

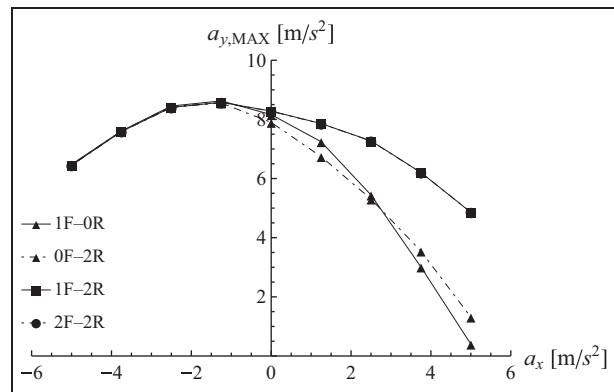


Figure 15. $a_{y,MAX}$ as a function of a_x at 90 km/h for the vehicle layouts 1F-0R, 0F-2R, 1F-2R and 2F-2R without torque-vectoring control (even left-to-right torque distribution).

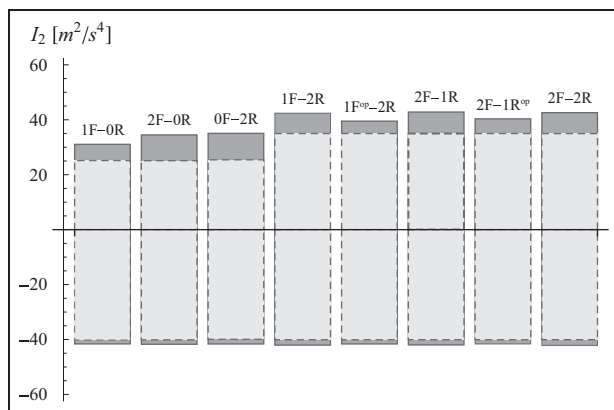


Figure 16. I_2 for the eight alternative electric vehicle layouts at 90 km/h without (light grey) and with (dark grey) torque-vectoring control.

control, a fixed front-to-rear distribution (50:50 in traction and 75:25 in braking) was used. Figure 16 shows the corresponding values of I_2 , indicating the potential performance in terms of limit cornering behaviour. The uncontrolled 4WD layouts present a significant

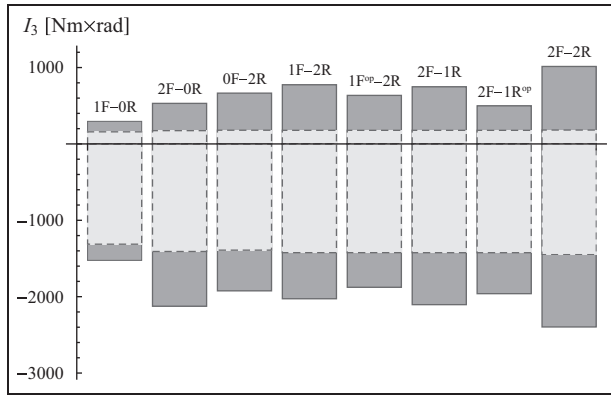


Figure 17. $I_{3,Stab}$ (negative values) and $I_{3,Dest}$ (positive values) for the eight alternative electric vehicle layouts, for a constant velocity (90 km/h), without (light grey) and with (dark grey) torque-vectoring control.

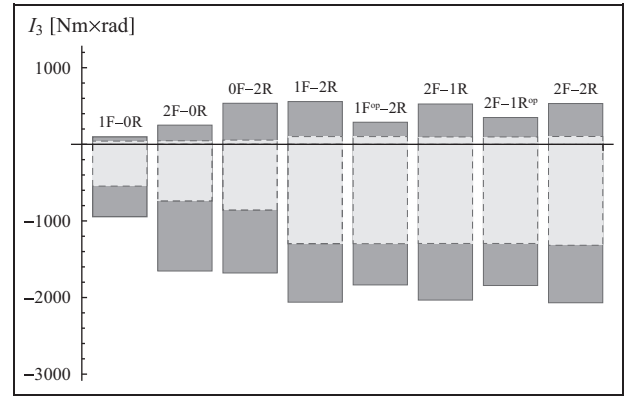


Figure 18. $I_{3,Stab}$ (negative values) and $I_{3,Dest}$ (positive values) for the eight alternative electric vehicle layouts, for a constant longitudinal acceleration ($a_x = 3 \text{ m/s}^2$), at 90 km/h, without (light grey) and with (dark grey) torque-vectoring control.

advantage over the uncontrolled 2WD layouts, i.e. approximately 27% in traction. By introducing torque vectoring, the 2WD layouts can reach the lateral acceleration limits of the uncontrolled 4WD layouts, as shown by the same I_2 values. With the highest I_2 values, the 4WD layouts with torque-vectoring control on both axes have the greatest capabilities in terms of limit cornering behaviour. Compared with the 4WD layouts with an open differential on the axle with a single motor drive, torque-vectoring on two axes allows an increase in I_2 of 6% in traction. With respect to the 2WD layouts, the possible operating area during traction increases by 20–27%, implying a significant extension of the limit handling region. For vehicle 1F-0R, the improvement in the cornering performance achievable with torque-vectoring control is marginal owing to limitation of the differential torque bias.

Figure 17 and Figure 18 plot $I_{3,Stab}$ and $I_{3,Dest}$ for vehicles at a constant velocity and vehicles at $a_x = 3 \text{ m/s}^2$ respectively. The layouts consisting of three motors and an open differential on one axle (1F^{op}-2R and 2F-1R^{op}) show a benefit in terms of controllability over the 2WD configurations only for vehicles without torque-vectoring control. When torque-vectoring control is considered, rear-wheel-drive vehicles are more effective than front-wheel-drive vehicles (vehicles 1F-0R and 2F-0R) and layouts with three electric motors and an open differential on one axle. In contrast, 4WD vehicle layouts with torque-vectoring control on both axes consistently provide an enhanced stabilisation capability. The vehicles with torque-vectoring differentials show lower stabilisation capabilities for dynamic steering inputs with respect to the other vehicle architectures as a consequence of the limitation imposed by the differential clutches on the torque bias allowable for torque-vectoring control.

In summary, as expected, the 2F-2R configuration yields the highest-performing vehicle in terms of the cornering behaviour. However, on the basis of the desired specifications, the designer can easily identify other suitable configurations as shown above. Once the

vehicle architecture has been selected, the engineer can re-apply the procedure with the constraints and the objective functions defined in the third section. The outcome of this step is the feedforward contribution of the controller that can be implemented, e.g. for further simulation studies in the time domain or potentially already on a prototype.

Conclusions

A novel methodology for the design of the feedforward part of the torque-vectoring controller of an FEV based on an experimentally validated quasi-static model was reported. Also, new vehicle-handling performance indicators were proposed for evaluation and comparison of the torque-vectoring potential of different electric drive-train layouts. In general, the simulation results obtained reveal the following.

1. The quasi-static model formulation is well suited to the quick assessment of the full range of vehicle-handling capabilities;
2. The offline optimisation procedure allows quick and precise development of the feedforward part of the torque-vectoring controller for a desired set of reference understeer characteristics;
3. The proposed indicators facilitate an objective comparison of the cornering behaviours of a vehicle in the linear and non-linear regions, including the yaw moment controllability.

With respect to the particular case-study vehicle data set, the results obtained show the following.

1. The number of driven axles, and not their locations, are the main relevant factor determining the extension of the area of linear vehicle response, which is the main perceivable parameter of vehicle-handling performance for normal drivers;

2. The 4WD layouts permit a tangible benefit over 2WD layouts in terms of the three handling performance indicators only when they allow torque-vectoring on both axles.

Declaration of conflict of interest

The authors declare that there is no conflict of interest.

Funding

This work was supported by the European Union Seventh Framework Programme FP7/2007-2013 under the E-VECTOORC project (grant number 284708).

References

1. Khaligh A and Li Z. Battery, ultracapacitor, fuel cell, and hybrid energy storage systems for electric, hybrid electric, fuel cell, and plug-in hybrid electric vehicles: state of the art. *IEEE Trans Veh Technol* 2010; 59(6): 2806–2814.
2. Zeraoulia M, El Hachemi Benbouzid M and Diallo D. Electric motor drive selection issues for HEV propulsion systems: a comparative study. *IEEE Trans Veh Technol* 2006; 55(6): 1756–1764.
3. Rinderknecht S and Meier T. Electric power train configurations and their transmission systems. In: *International symposium on power electronics, electrical drives, automation and motion*, Pisa, Italy, 14–16 June 2010, pp. 1564–1568. New York: IEEE.
4. Hoehn BR, Stahl K, Gwinner P and Wiesbeck F. Torque-vectoring driveline for electric vehicles. In: *FISITA world automotive congress*, Lecture Notes in Electrical Engineering, Vol 191, Beijing, People's Republic of China, 27–30 November 2012, pp. 585–593. Berlin: Springer.
5. De Novellis L, Sorniotti A, Gruber P et al. Torque-vectoring for electric vehicles with individually controlled motors: state-of-the-art and future developments. In: *26th electric vehicle symposium*, Los Angeles, California, USA, 6–9 May 2012. Red Hook, New York: Curran Associates. D 7.2.
6. De Novellis L, Sorniotti A and Gruber P. Optimal wheel torque distribution for a four-wheel-drive fully electric vehicle. SAE paper 2013-01-0673, 2013.
7. Crolla DA and Cao D. The impact of hybrid and electric powertrains on vehicle dynamics, control systems and energy recuperation. *Veh System Dynamics* 2012; 50(Suppl 1): 95–109.
8. van Zanten A, Erhardt R and Pfaff G. VDC, the vehicle dynamics control system of Bosch. SAE paper 950749, 1995.
9. Goodarzi A and Alirezaie M. A new fuzzy-optimal integrated AFS/DYC control strategy. In: *8th international symposium on advanced vehicle control*, Taipei, Republic of China, 20–24 August 2006, paper AVEC060050, pp. 65–70. Tokyo: JSAE.
10. Nagai M, Shino M and Gao F. Study on integrated control of active front steer angle and direct yaw moment. *JSAE Rev* 2002; 23(3): 309–315.
11. Abe M. A Theoretical analysis on vehicle cornering behaviors in acceleration and in braking. *Veh System Dynamics* 1986; 15(Suppl 1): 1–14.
12. Nagai M, Hirano Y and Yamanaka S. Integrated control of active rear wheel steering and direct yaw moment control. *Veh System Dynamics* 1997; 27(5–6): 357–370.
13. Shimada K and Shibahata Y. Comparison of three active chassis control methods for stabilizing yaw moments. SAE paper 940870, 1994.
14. van Zanten A. Bosch ESP systems: 5 years of experience. SAE paper 2000-01-1633, 2000.
15. Canale M. Vehicle yaw control via second-order sliding-mode technique. *IEEE Trans Ind Electron* 2008; 55(11): 3908–3916.
16. Chang S and Gordon TJ. Model-based predictive control of vehicle dynamics. *Int J Veh Autonomous Systems* 2007; 5(1–2): 3–27.
17. Koibuchi K, Yamamoto M, Fukada Y and Inagaki S. Vehicle stability control in limit cornering by active brake. SAE paper 960487, 1996.
18. Hattori Y. Optimum vehicle dynamics control based on tyre driving and braking forces. *R&D Toyota CRDL* 2003; 38(4): 23–29.
19. Graf M and Lienkamp M. Torque vectoring control design based on objective driving dynamic parameters. In: *FISITA world automotive congress*, Lecture Notes in Electrical Engineering, Vol 196, Beijing, People's Republic of China, 27–30 November 2012, pp. 1787–1796. Berlin: Springer.
20. ISO 8855:2011 *Road vehicles – vehicle dynamics and road-holding ability – vocabulary*. Geneva: International Organization for Standardization, 2011.
21. Sawase K, Ushiroda Y and Miura T. Left–right torque-vectoring technology as the core of super all wheel control (S-AWC). *Mitsubishi Motors Tech Rev* 2006; 18: 18–24.
22. Gao L. Dynamic lithium-ion battery model for system simulation. *IEEE Trans Components Packag Technol* 2002; 25(3): 495–505.
23. IPG Automotive. CarMaker®, <http://www.ipg.de/Car-Maker.609.0.html> (2013, accessed 25 April 2013).
24. Amann N, Böcker J and Prenner F. Active damping of drive train oscillations for an electrically driven vehicle. *IEEE/ASME Trans Mechatronics* 2004; 9(4): 697–700.
25. ISO 4138:2012 *Passenger cars – steady-state circular driving behaviour – open-loop test methods*. Geneva: International Organization for Standardization, 2012.
26. ISO 7401:2011 *Road vehicles – lateral transient response test methods – open-loop test methods*. Geneva: International Organization for Standardization, 2011.
27. Milliken WF and Milliken DL. *Race car vehicle dynamics*. Warrendale, Pennsylvania: SAE International, 1995.
28. Shibahata Y, Shimada K and Tomari T. Improvement of vehicle maneuverability by direct yaw moment control. *Veh System Dynamics* 1993; 22(5–6): 465–481.
29. Tanaka S, Kawamoto M and Inagaki H. *Regenerative braking electric vehicle with four motors*. US Patent 5148883, 1992.
30. Genta G. *Motor vehicle dynamics: modelling and simulation*. Singapore: World Scientific Publishing, 1997.
31. Alexander MJ, Allison JT and Papalambros PY. Decomposition-based design optimisation of electric vehicle powertrains using proper orthogonal decomposition. *Int J Powertrains* 2011; 1(1): 72–92.
32. Coleman TF and Li Y. An interior trust region approach for nonlinear minimization subject to bounds. *SIAM J Optimization* 1996; 6(2): 418–445.
33. Sawase K and Ushiroda Y. Improvement of vehicle dynamics by right-and-left torque-vectoring system in various drivetrains. *Mitsubishi Motors Tech Rev* 2007; 20:14–20.

Appendix I

Notation

a_x	longitudinal acceleration of the vehicle	J_P	objective function related to the energy efficiency
a_{x_i}	component of the wheel hub acceleration in the longitudinal direction of the wheel reference system	J_{W_i}	moment of inertia of the wheel
a_y	lateral acceleration of the vehicle	J_{l_i}	moment of inertia of the primary transmission shaft
a_y^*	lateral acceleration threshold for a linear cornering response	J_{2_i}	moment of inertia of the secondary transmission shaft
$a_{y,MAX}$	maximum value of the lateral acceleration of the vehicle in cornering	J_{3_i}	moment of inertia of the output transmission shaft
d_{CG}	height of the roll axis measured at the centre of gravity of the vehicle	K_U	understeer gradient = $\partial\delta/\partial a_y$
d_F	height of the roll axis measured at the front suspension	L	wheelbase of the vehicle
d_R	height of the roll axis measured at the rear suspension	m	overall mass of the vehicle
f_0	constant term of the rolling resistance of the wheel	n_{mot}	number of electric motor drives
f_1	linear term of the rolling resistance of the wheel	M_z	yaw moment
f_2	quadratic term of the rolling resistance of the wheel	M_{z,F_x}	yaw moment contribution due to the longitudinal forces of the tyre
F_{drag}	aerodynamic drag resistance in the longitudinal direction of the vehicle	M_{z,F_y}	yaw moment contribution due to the lateral forces of the tyre
F_{x_i}	tyre–road contact force in the longitudinal direction of the tyre reference system	M_{z_i}	self-aligning moment at the tyre–road contact
F_{y_i}	tyre–road contact force in the lateral direction of the tyre reference system	M_{φ_F}	front anti-roll moment
F_{z_i}	vertical force of the tyre	M_{φ_R}	rear anti-roll moment
F_{z0_i}	static vertical force of the tyre	p_i^{ref}	reference friction brake pressure
h_{CG}	height of the centre of gravity of the vehicle	p_i	friction brake pressure
I_1	performance parameter indicating the extension of the linear cornering characteristics of the vehicle	P_{input_i}	input power of a motor drive
I_2	performance parameter indicating the achievable limits of the cornering characteristics of the vehicle	r	yaw rate
$I_{3, Dest}$	performance parameter indicating the destabilising contribution of the yaw moment diagram	r_{ref}	reference yaw rate
$I_{3, Stab}$	performance parameter indicating the stabilising contribution of the yaw moment diagram	R_i	laden radius of the tyre
J_A	objective function related to the maximum lateral acceleration	R_{roll_i}	rolling radius of the tyre
J_{eq_i}	equivalent moment of inertia of the powertrain at the motor side	T_{dLeft}	left output torque of the torque-vectoring differential
J_{HS_i}	moment of inertia of the half-shaft	T_{dRight}	right output torque of the torque-vectoring differential
J_{mot_i}	moment of inertia of the electric motor	T_{FB_i}	friction brake torque
J_M	objective function related to the maximum and minimum yaw moments	T_i	drivetrain torque at the wheel
		T_{m_i}	torque due to the electric motor
		$T_{m_i}^{ref}$	reference torque due to the electric motor
		T_{tot}^c	total reference torque due to the wheels
		T_{TOT}	sum of the wheel torques
		u	longitudinal component of the speed of the vehicle in the vehicle reference system
		V	modulus of the velocity vector of the vehicle
		V_{x_i}	component of the velocity of the wheel hub in the longitudinal direction of the tyre reference system
		x_i	longitudinal coordinate in the vehicle reference system
		y_i	lateral coordinate in the vehicle reference system
		α_i	slip angle
		β	side-slip angle of the vehicle
		γ_i	camber angle
		δ	steering-wheel angle
		δ_{w_i}	steering angle of the wheel

$\eta_{CV1_i}, \eta_{CV2_i}$	efficiencies of the constant-velocity joints	τ_{1_i}	gear ratio of the first reduction stage
η_{1_i}	efficiency of the first reduction stage	τ_{2_i}	gear ratio of the second reduction stage
η_{2_i}	efficiency of the second reduction stage	ω_i	angular speed of the wheel
φ	roll angle	ω_{m_i}	angular speed of the electric motor
σ_i	slip ratio coefficient in the longitudinal direction of the tyre reference system	.	time derivative of a variable

Appendix 2

Main vehicle parameters

A list of the main vehicle parameters for the six layouts in Figure 1 is given in Table 1.

Table 1. List of the main vehicle parameters.

Mass (kg)	2255 (vehicle 1F-0R) 2270 (vehicles 2F-0R and 0F-2R) 2295 (vehicles 1F-2R and 2F-1R) 2300 (vehicle 2F-2R)
Mass distribution (% front:% rear)	61:39
Front track (m)	1.625
Rear track (m)	1.625
Wheelbase (m)	2.59
Aerodynamic drag coefficient (-)	0.339
Frontal area (m ²)	2.032
Wheel radius (m)	0.3705
f_0 (-)	0.012
f_1 (s/m)	0
f_2 (s ² /m ²)	6.5×10^{-6}
Peak torque of the motor(s) (N m)	908 (vehicle 1F-0R) 454 (vehicles 2F-0R and 0F-2R) 454 and 227 (front and rear motors respectively of vehicle 1F-2R) 227 and 454 (front and rear motors respectively of vehicle 2F-1R) 227 (vehicle 2F-2R)
Base speed of the motor(s) (r/min)	4838
Overall drivetrain gear ratio (-)	10